Get Started with Generative AI & Agents in Azure

Microsoft Azure and Microsoft Foundry enable you to build intelligent applications using generative AI models and autonomous agents — from experimentation to production.

AZURE AI | MICROSOFT FOUNDRY | GENERATIVE AI

Introduction to Generative AI & Agents

Generative AI creates new content — text, images, code — transforming industries. **Agents** pair models with instructions and tools to automate real workflows.

Content Creation

Blogs, marketing copy, and media

Customer Support

AI chatbots for 24/7 assistance

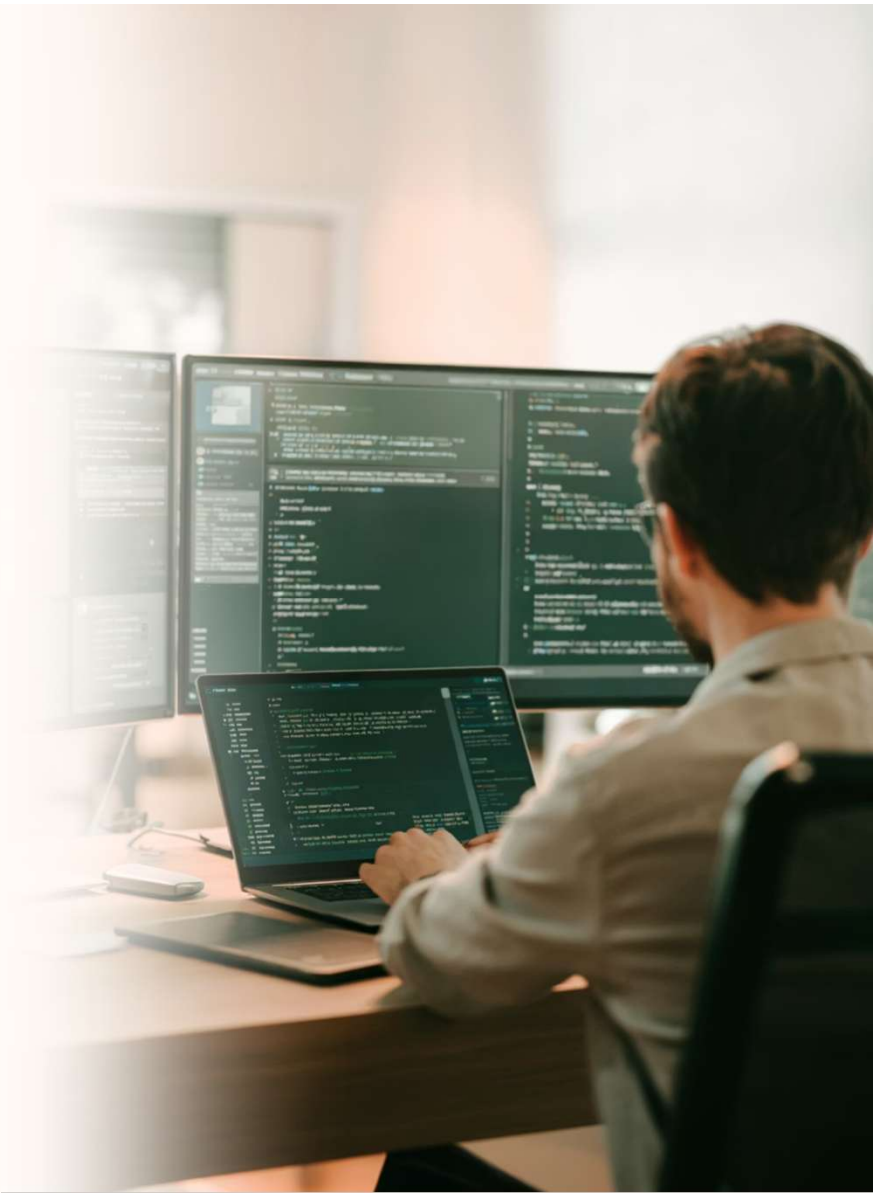
Code Generation

Developer productivity tools

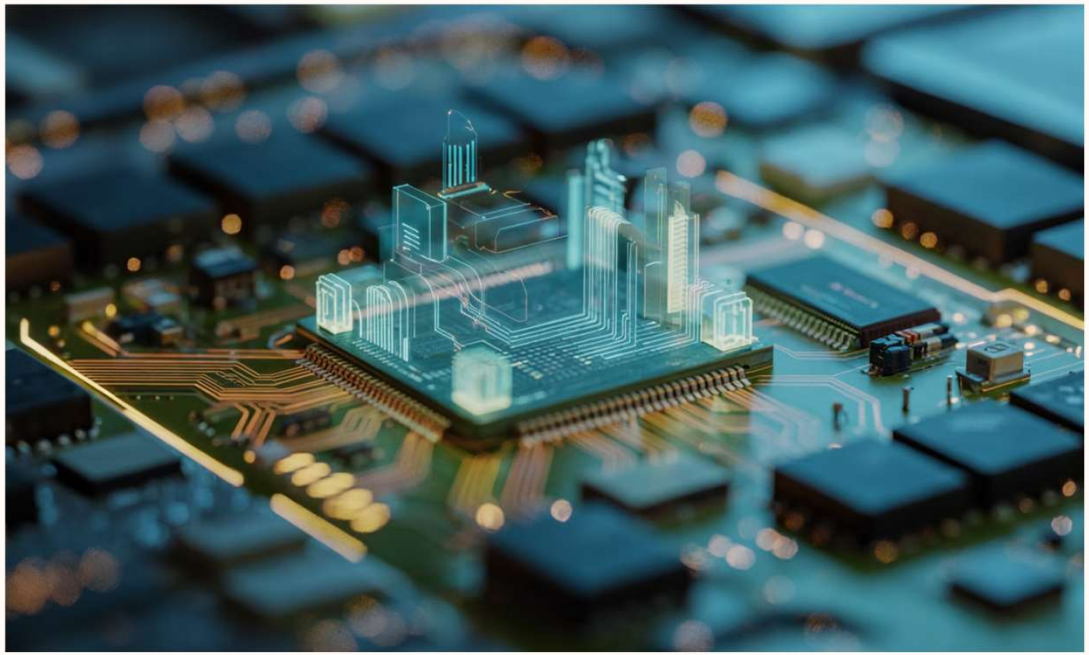
Personalized Learning

Adaptive educational experiences

Key Concept: Agents combine **model + instructions + tools** into reusable, consistent AI solutions.



Generative AI Models & Microsoft Foundry



Microsoft Foundry

A unified platform to **discover, evaluate, and deploy** AI models — supporting copilots, agents, and full AI applications.

Model Catalog

Central hub for Azure, partner, and open-source models with benchmarks, fine-tuning options, and Responsible AI docs.

Popular Model Families

- **GPT-5.x** — Advanced reasoning
- **Claude Opus 4.5** — Complex coding & agents
- **Mistral Large 3** — Balanced performance
- **GPT-4.1** — Fast, real-time apps

Evaluation & Deployment

Evaluation Metrics

Assess model quality before deploying to production.

Standard Metrics

Accuracy, Precision, Recall, F1

AI Quality Metrics

Relevance, Coherence, Fluency

Tools

Benchmarks, leaderboards, safety evaluators

Deployment via API

Models deploy as scalable, secure API endpoints.

- **Tokens** — Smallest unit of text processed
- **TPM** — Tokens Per Minute, your processing capacity
- **Throttling** — Rate limits prevent overload
- Configurable: deployment type, model version, TPM

Outcome: Scalable, secure, production-ready AI systems.

Using the Foundry Playground

The Playground is your sandbox — test prompts, tune parameters, and auto-generate API code before writing a single line.

Temperature

Controls creativity vs. consistency. Higher = more varied; lower = more deterministic.

Max Output Tokens

Limits response length and cost. Set based on expected output size.

System Instructions

Define the assistant's role, tone, and guardrails. Shapes every response.

Prompt Types

User Prompt = the query. **System Prompt** = role & rules. Reuse both directly in code.

Chat Clients, Models & Agents

Lightweight Chat Client

A minimal CLI or web app that sends user input to a model via API — keeping compute server-side. Easy to build, test, and extend.

Python Integration

- Uses OpenAI-compatible API
- Connects via endpoint + API key
- Sends prompts, receives structured responses

Models vs. Agents

Models provide raw intelligence — they generate output from input.

Agents are task-oriented systems built *on top of* models, orchestrating multi-step workflows.

Playground → Experimentation **API** → Real-world applications

Creating & Using Agents in Microsoft Foundry

Agents are reusable AI applications that break complex tasks into steps, maintain memory, and call external tools automatically.

Model
Reasoning engine (e.g., GPT-4.1)

Instructions
Role, tone, and operational constraints

Tools
APIs, functions, and retrieval systems

Every agent is built from these three pillars — together they enable consistent, workflow-driven AI behavior.

Break Tasks into Steps
Decompose complex goals automatically

Conversation Memory
Maintain context across turns

External Tool Calls
Invoke APIs and functions autonomously

Structured Outputs
Generate consistent, parseable results

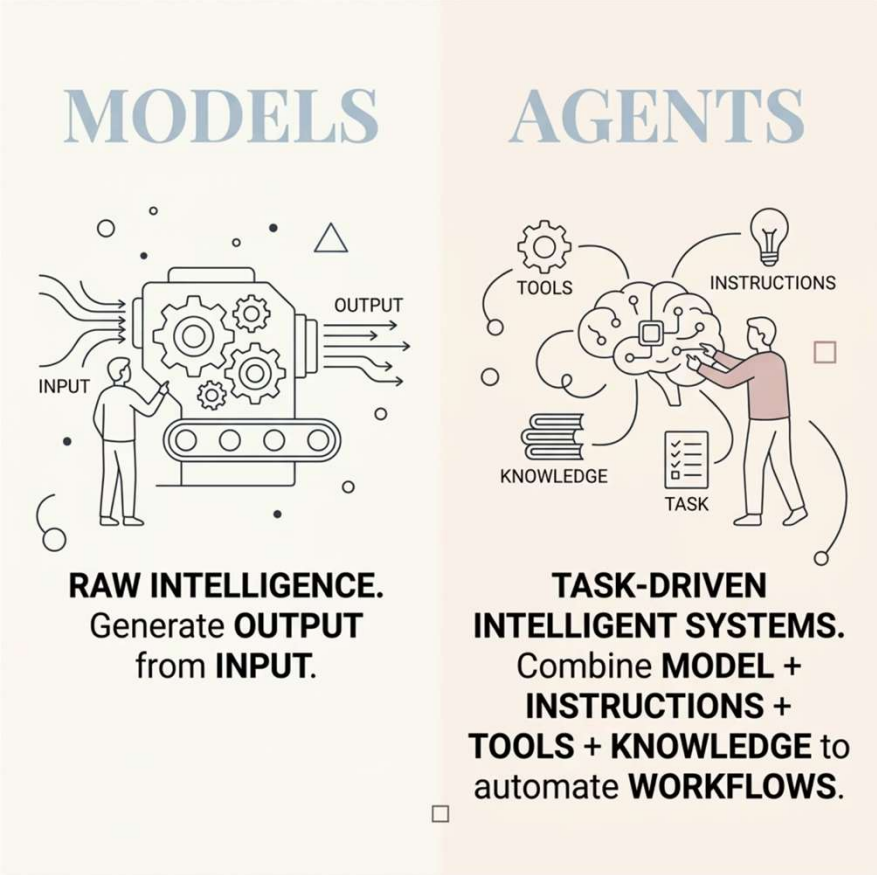
Tools, Knowledge & Publishing Agents

Tools & Knowledge

- **Tools** = Actions — APIs, code execution, workflows
- **Knowledge** = Context — documents and datasets via RAG
- Grounded responses improve accuracy and reduce hallucination

Using & Publishing Agents

- Access via **Project API** using agent-id and Foundry SDK (Python)
- Integrate into apps, bots, and automated workflows
- Publishing converts the agent into a **managed Azure resource**
- Provides a secure, stable endpoint for production and team sharing



Summary: Your Generative AI Journey

Microsoft Foundry supports the **full lifecycle** — from model discovery to production-ready agentic applications.

01

Language Models (LLMs)

The foundation — text generation, coding, reasoning, multimodal tasks

02

Model Catalog

Discover and compare models from Microsoft, partners, and open-source

03

Deployment

Expose models as scalable, secure API endpoints with configurable TPM

04

Client Applications

Lightweight apps and Python scripts using the Foundry SDK

05

Agents

Combine model + instructions + tools + knowledge into reusable, workflow-driven AI

Outcome: Build secure, scalable, production-ready AI solutions — the foundation for modern generative and agentic applications.